

Task12-Missing value Identification

1. Introduction

This project was completed as part of Task 12 of the Veda Technology Data Analytics Internship. The purpose of the task is to identify missing values in a dataset, measure how frequently they occur, and summarize the columns affected by missing data. The analysis was performed using Python and Pandas with the Titanic dataset.

2. Objective

The main objective of this task is to perform basic missing-data inspection. The project focuses on counting null values by column, calculating the percentage of missing values, creating a summary table, and documenting suitable considerations for handling the missing data.

3. Dataset Used

The Titanic training dataset was used for this task. It contains 891 passenger records and includes information such as passenger class, name, sex, age, number of siblings/spouses, number of parents/children, ticket, fare, cabin, and port of embarkation.

4. Tools and Technologies

Python 3.11

Pandas

Jupyter Notebook

Microsoft Excel

5. Methodology

Loaded the Titanic dataset using Pandas.

Checked the dataset shape and basic information.

Used `df.isnull().sum()` to count missing values in each column.

Calculated the percentage of missing values for each column.

Filtered the result to show only columns containing missing values.

Sorted the summary from the highest to the lowest number of missing values.

Exported the missing-value summary for viewing in Excel.

Did not remove rows or columns because the task is focused on identification and justified analysis.

6. Missing Value Summary

The analysis identified missing values in three columns: Cabin, Age, and Embarked.

Column	Missing Values	Missing Percentage
Cabin	687	77.10%

Age	177	19.87%	
Embarked	2	0.22%	

7. Findings and Analysis

7.1 Cabin

The Cabin column contains 687 missing values, representing 77.10% of the dataset. This is a very high level of missing data. Rows with missing Cabin values should not be deleted blindly because doing so would remove a large portion of the dataset.

7.2 Age

The Age column contains 177 missing values, representing 19.87% of the dataset. This is a moderate amount of missing data. If Age is required for further analysis, methods such as median imputation or group-based imputation could be considered. No imputation was performed because this task focuses on identification and summary.

7.3 Embarked

The Embarked column contains 2 missing values, representing 0.22% of the dataset. These records can be inspected individually before selecting an appropriate method for handling them.

8. Interview Questions

Q1. What is a missing value?

A missing value is a data field for which no valid value has been recorded. In Pandas, missing values are commonly represented by NaN or None.

Q2. Why can blindly deleting rows be risky?

Blindly deleting rows can reduce the size of the dataset and may remove useful information. If missing values are concentrated in a particular group, deleting those records can also affect the representativeness of the data and potentially introduce bias.

9. Conclusion

The missing-value identification task was successfully completed using Python and Pandas. The analysis found that Cabin has the largest amount of missing data, followed by Age and Embarked. A missing-value summary was created and exported for further use. No rows or columns were deleted without justification. The results provide a basis for making appropriate data-cleaning decisions in later analysis. Use this info and make report